

THERMACART · REQUIREMENTS

ThermaCart requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from TCT-CAL-001; R12 oven temperature corrected to the RT 70 HC limit; R16 budget scope per TCT-DDR-001 D6
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

ThermaCart requirements

These are the requirements for the TC-L cartridge and its adapter frame. The targets were proposed at TRL 2, and the design choices behind them were decided by Amish on 2026-09-25 (TCT-DDR-001 and TCT-DDR-002). Under TCT-DDR-002 the C25 band of R1 is widened to 22 to 27 °C (O9), R5 is relaxed to 38 % for the stock-tube build with 50 % kept as the target for a later extrusion (O4), and R14 carries a labelled handling rule for H70 cartridges taken from an oven (O7). R10 keeps its 8 h target (O5). The status column comes from TCT-CAL-001 v0.2, which uses the decided dark finish (O8). Two requirements are **not met** on paper (R10 and R16), one is **at risk** (R12), three can only be settled by tests (R6, R7 and R17) and twelve are met.

Table 1. Requirements

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (TCT-CAL-001)
R1	Offer a small set of standard temperature grades	At least three: C5 (melt 4 to 6 °C, cold chain), C25 (22 to 27 °C, comfort; widened from 24 to 27 °C by TCT-DDR-002, O9), H70 (68 to 72 °C, hot holding)	Material datasheets; later DSC test of each fill batch	Met: RT 5 HC 5 to 6 °C, RT 25 HC 22 to 26 °C, RT 70 HC 69 to 71 °C
R2	Common envelope that fits kitchen and cooler formats	Fits a GN 1/3 slot: at most 325 x 176 mm footprint and 65 mm high, handle included	Model and drawing	Met: 323.4 x 152.4 x 63.5 mm with the bail stowed (TCT-DWG-001)
R3	Enough stored energy per cartridge	60 Wh (216 kJ) or more of usable storage within ± 3 K of the melt point	Calculation per grade	Met: C5 65.4 Wh, C25 60.5 Wh (thin margin), H70 72.9 Wh
R4	Carry with one hand	Filled mass 3.0 kg or less	Mass estimate, then weighing	Met, thin margin: 2.90 to 2.98 kg
R5	Efficient use of mass	PCM at least 38 % of filled mass for the stock-tube build; 50 % remains the target for a later custom extrusion (relaxed by TCT-DDR-002, O4)	Mass estimate	Met, thin margin: 38.1 to 39.7 %
R6	Sealed for life	No leak through 1,000 thermal cycles and a 1 m drop onto concrete in any state	Seal design review at TRL 3; cycle and drop tests belong to TRL 4	Not verifiable at TRL 3; an end-on drop of a molten cartridge could see a surge of up to 5 bar
R7	Long life	1,000 cycles or more with less than 10 % loss of latent capacity	Literature and supplier data at TRL 3	Not verifiable at TRL 3; plausible for paraffins
R8	No pressure hazard	Internal gauge pressure 0.5 bar or less at the highest charge temperature; ullage of at least 10 % of the inner volume at that temperature	Calculation	Met: 0.00 bar at the fill temperature, +0.29 bar at 20 K above the limit; wall stress 56 MPa under the cold vacuum
R9	Wrong grade cannot be fitted	Each grade has its own nose key; a frame for one grade rejects the others	Design review of key geometry	Met by design: every pairing checked
R10	Cold-chain hold time	Two C5 cartridges keep a 25 L cooler (heat leak about 0.5 W/K, estimate) at 2 to 8 °C for 8 h or more at 32 °C ambient	Heat balance calculation	Not met: 2.4 h with the decided dark finish (1.9 h bare); the stored energy would last 9.7 h, but air-side heat transfer limits the hold. Produce laid directly on the cartridges (TCT-DDR-002, O5) is not credited and waits for a TRL 4 measurement, which is on hold
R11	Hot-holding time	One H70 cartridge keeps food in an insulated GN carrier (about 0.25 W/K,		

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (TCT-CAL-001)
		estimate) at 63 °C or more for 4 h or more at 25 °C ambient	Heat balance calculation	Met: 13.6 h with the dark finish (7.2 h bare)
R12	Recharge with common equipment	Full recharge in 8 h or less: C5 and C25 in a domestic freezer or refrigerator, H70 in an oven set to 85 °C (never above 90 °C) or on a 60 W heating pad	Calculation (Stefan problem plus fin convection)	At risk: with the dark finish, C5 freezer 4.2 h, C25 refrigerator 4.6 h, H70 oven 5.3 h; H70 pad 2.7 to 10.3 h
R13	State of charge visible	Charged or melted state readable at a glance without tools	Design review	Met by design (sight window on a PCM vial)
R14	Safe to handle when charged	Handle touch temperature below the ISO 13732-1 burn threshold for 10 s contact on every grade; for an H70 cartridge taken from an oven, the label states a handling rule (oven gloves, or wait 12 min) and pad charging is the preferred method (TCT-DDR-002, O7)	Calculation of handle temperature; label review	Met with the handling rule: bail on a thermal break settles at 42.5 °C on pad charging; after an oven it is at 85 °C and falls below 55 °C in about 12 min, which the label rule covers
R15	Materials compatible with each fill	No corrosion, swelling or permeation: aluminium shell with FKM or NBR seals for paraffins; coated shell or polymer liner for salt hydrates	Compatibility review against supplier data	Met by design for paraffin grades (epoxy rated 120 °C or more); open for salt hydrates
R16	Low cost and garage-buildable	\$50 or less in parts per cartridge and \$20 or less per frame; stock sections, drill press and taps only. The \$80 prototype budget covers one C5 cartridge and one frame (TCT-DDR-001 D6)	Priced BOM	Not met: \$76 per cartridge including the dark finish; frame \$16 met; one cartridge and one frame \$92 against \$80 (indicative prices). Supplier quotes come first and the budget is not raised until they are in (TCT-DDR-002, O6)
R17	Open and documented	Envelope, key geometry, grade table and fill recipe published under CERN-OHL-S-2.0	Repository review	Not verifiable at TRL 3; envelope, key and grade tables are published, the fill recipe waits for TRL 4
R18	Recoverable at end of life	Fill can be drained or melted out through the fill port and the shell reused; fill named on the label	Design review	Met by design

■ Assumptions

- PCM data from the Rubitherm datasheets for RT 5 HC, RT 25 HC and RT 70 HC, taken at the low side of their ± 7.5 % tolerance; usable storage is the latent heat plus sensible heat within ± 3 K of the melt point (TCT-CAL-001, Table 1).
- Fill of 1.10 to 1.18 kg: 90 % of a 1.655 L inner volume, filled as liquid at each grade's maximum operating temperature (45, 65 and 90 °C).

- Hot-holding threshold of 63 °C follows common food-safety practice (the US FDA Food Code uses 57 °C, 135 °F); the lower value would lengthen the hold time.
- Cooler and carrier heat leaks are estimates for typical polyurethane and EPP boxes and must be replaced with measured values from a partner's real boxes.
- The burn thresholds used for R14 (about 55 °C for bare metal and 70 °C for plastics, 10 s contact) are commonly quoted from ISO 13732-1 and have not been checked against the standard.